22050 Continuous-Time Signals and Linear Systems

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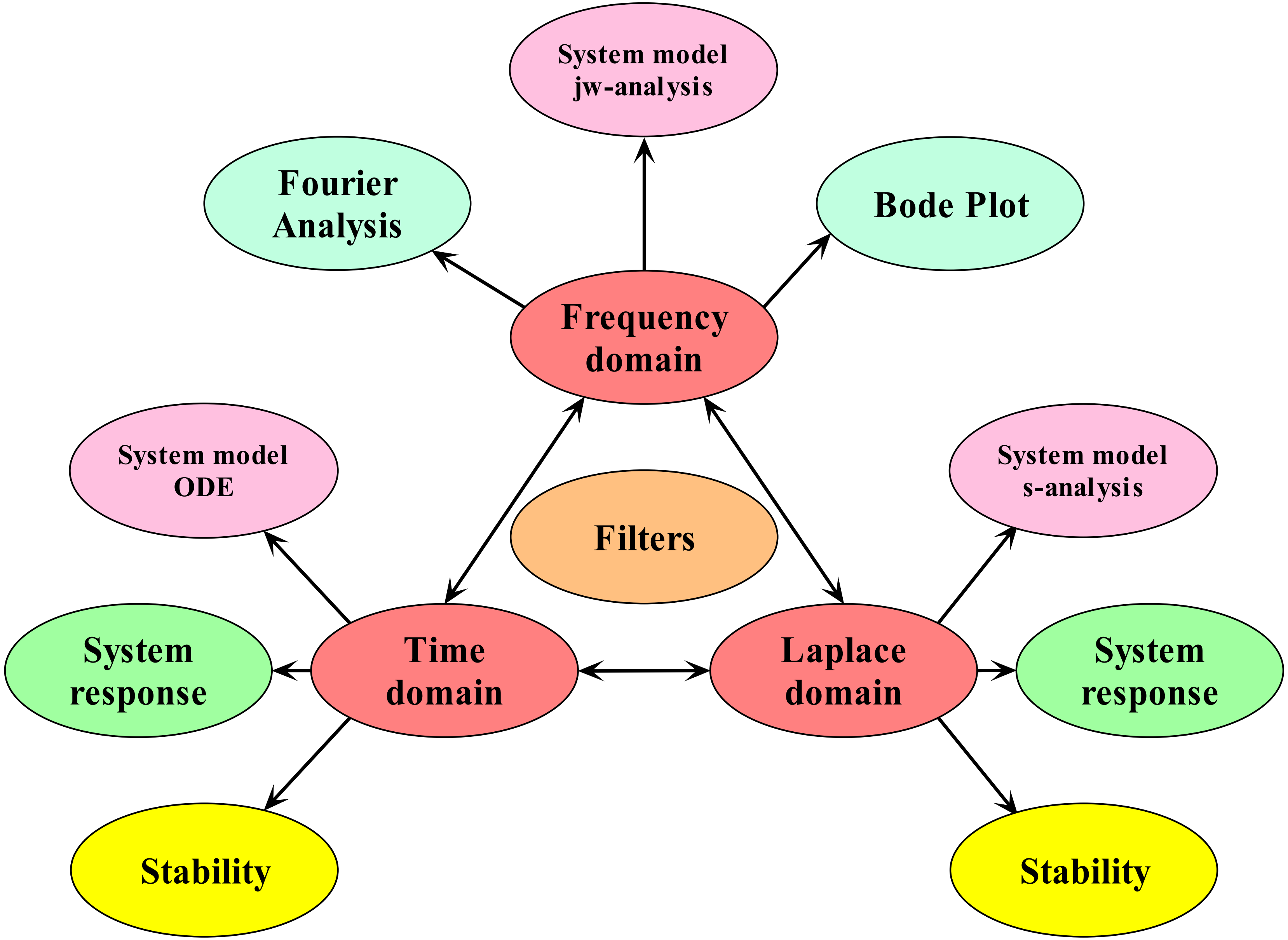
L04

Expansion of signals using orthogonal functions.
Expansion and reconstruction of signals using Fourier series.

Course overview

- System
  - Model
  - Response
  - Performance

- Domains
  - Time
  - Frequency
  - s-domain



Lecture plan

- Signal representation 3.1 – 3.3
  - Components of signals
  - Signal correlation and orthogonality
  - Orthogonal signal space
- Trigonometric Fourier Series 3.4 – 3.5
  - Orthogonality of trigonometric basis functions
  - Trigonometric Fourier Series.
  - Compact trigonometric Fourier Series.
  - Conditions for existence.
  - Symmetry in even and odd signals
- Complex exponential Fourier Series 3.6
  - Orthogonality of complex exponential basis functions.
  - Determining coefficients
  - Symmetry properties
  - Negative frequencies
  - Power spectrum

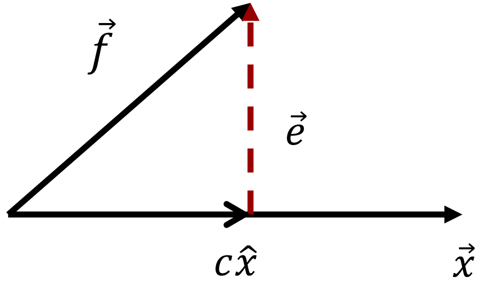
Lathi: Ch. 3.1 – 3.3 incl.

Similarity

Video 1

Components of a vector and signals

We approximate a vector $\vec{f}$ by another vector $c\hat{x}$, where $|\hat{x}| = 1$.



$$\vec{f} \approx c\hat{x}$$

The difference defines the error:

$$\vec{e} = \vec{f} - c\hat{x}$$

The length of the error vector indicates the magnitude of the error.

$$|\vec{e}|^2 = \vec{e}\cdot\vec{e} = (\vec{f} - c\hat{x})\cdot(\vec{f} - c\hat{x})$$
$$= (\vec{f}\cdot\vec{f}) + c^2(\hat{x}\cdot\hat{x}) - 2c(\vec{f}\cdot\hat{x})$$

$$\frac{d(|\vec{e}|^2)}{dc} = 0 + 2c|\hat{x}|^2 - 2(\vec{f}\cdot\hat{x}) = 0 \Rightarrow$$

We want to choose c so that the error is the smallest possible:

$$\boxed{c = \frac{\vec{f}\cdot\hat{x}}{|\hat{x}|^2} = \vec{f}\cdot\hat{x}}$$

Signal space

$$f(t) \approx cx(t), \qquad t_1 \le t \le t_2$$

$$e(t) = \begin{cases} f(t) - cx(t) & t_1 \le t \le t_2 \\ 0 & \text{otherwise} \end{cases}$$

We optimize the approximation by minimizing the energy of the error signal:

$$E_e = \int_{t_1}^{t_2} e^2(t)dt = \int_{t_1}^{t_2} [f(t) - cx(t)]^2 dt$$

$$\frac{dE_e}{dc} = 0 \Rightarrow$$

$$\boxed{c = \frac{1}{E_x}\int_{t_1}^{t_2} f(t)x^*(t)dt}$$

Derivation on next two slide not lectured. Read for your self.

Mathematical elaboration (not lectured)

We will elaborate on this briefly.

$$E_e = \int_{t_1}^{t_2} e^2(t)dt = \int_{t_1}^{t_2} [f(t) - cx(t)]^2 dt$$

Assuming functions to be complex-valued:

$$|u+v|^2 = (u+v)(u^*+v^*) = |u|^2 + |v|^2 + u^*v + uv^*$$

$$u(t) \stackrel{\text{def}}{=} f(t) \qquad v(t) \stackrel{\text{def}}{=} -c\,x(t)$$

$$|f-cx|^2 = (f-cx)(f^*-cx^*) = |f|^2 + c^2|x|^2 - f^*cx - fcx^*$$

We aim to split the energy of the error into several terms, some of which will include the scaling factor c:

$$E_e = \int_{t_1}^{t_2} (|f|^2 + c^2|x|^2 - f^*cx - fcx^*)\,dt$$

$$= \int_{t_1}^{t_2} |f|^2\,dt + c^2 \int_{t_1}^{t_2} |x|^2\,dt - c\int_{t_1}^{t_2} f^*x\,dt - c\int_{t_1}^{t_2} fx^*\,dt$$

$$= E_f + c^2E_x - c\int_{t_1}^{t_2} f^*x\,dt - c\int_{t_1}^{t_2} fx^*\,dt$$

We want to determine the value of c which minimizes the energy of the error signal.

Mathematical elaboration (not lectured)

$$E_e = E_f + c^2E_x - c\int_{t_1}^{t_2} f^*x\,dt - c\int_{t_1}^{t_2} fx^*\,dt + \frac{1}{E_x}\int_{t_1}^{t_2} fx^*\,dt\int_{t_1}^{t_2} f^*xdt - \frac{1}{E_x}\int_{t_1}^{t_2} fx^*\,dt\int_{t_1}^{t_2} f^*xdt$$

add and subtract

$$E_e = E_f - \left|\frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt\right|^2 + c^2E_x - c\sqrt{E_x}\frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} f^*x\,dt - c\sqrt{E_x}\frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt + \frac{1}{E_x}\int_{t_1}^{t_2} fx^*\,dt\int_{t_1}^{t_2} f^*xdt$$

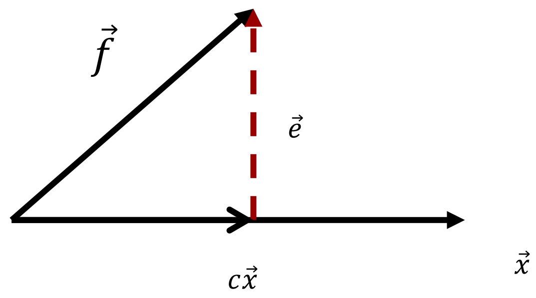
$$E_e = E_f - \left|\frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt\right|^2 + \left(c\sqrt{E_x} - \frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt\right)\left(c\sqrt{E_x} - \frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} f^*xdt\right)$$

complex conjugated

$$E_e = E_f - \left|\frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt\right|^2 + \left|c\sqrt{E_x} - \frac{1}{\sqrt{E_x}}\int_{t_1}^{t_2} fx^*\,dt\right|^2$$

$$\frac{dE_e}{dc} = 0 \Rightarrow \quad \boxed{c = \frac{1}{E_x}\int_{t_1}^{t_2} f(t)x^*(t)dt}$$

E_e becomes independent of c if the red term is zero.



Similarity of vectors:

$$c = \frac{\vec{f} \cdot \hat{x}}{|\hat{x}|^2} = \vec{f} \cdot \hat{x}$$

If c is large and positive, the two vectors are **similar**.

If c is large and negative, the two vectors are **opposite**.

If c is zero, the two vectors have nothing in common. They are **orthogonal**.

A similar measure of similarity can be defined for signals:

$$c = \frac{1}{\sqrt{E_f E_x}} \int_{-\infty}^{\infty} f(t)x^*(t)dt$$

Maximum similarity: $c = 1$

Maximal opposite: $c = -1$

Nothing in common: $c = 0$

In this case the signals are **orthogonal**:

$$\boxed{\int_{-\infty}^{\infty} f(t)x^*(t)dt = 0}$$

Energy of a sum of orthogonal signals

Let $z(t)$ be the sum of two complex valued and orthogonal signals:

$$z(t) = x(t) + y(t) \qquad\qquad x(t) \perp y(t)$$

The energy of $z(t)$ is:

$$E_z = \int_{t_1}^{t_2} |z(t)|^2 dt = \int_{t_1}^{t_2} |x(t) + y(t)|^2 dt$$

$$= \int_{t_1}^{t_2} \big(x(t) + y(t)\big)\big(x^*(t) + y^*(t)\big) dt$$

$$= \int_{t_1}^{t_2} |x(t)|^2 dt + \int_{t_1}^{t_2} |y(t)|^2 dt + \underbrace{\int_{t_1}^{t_2} x(t)y^*(t) dt}_{=0\ (\perp)} + \underbrace{\int_{t_1}^{t_2} x^*(t)y(t) dt}_{=0\ (\perp)}$$

$$= E_x + E_y$$

Observation:

The energy is preserved if we expand a signal in components that are orthogonal.

Correlation, an application of similarity measure

We have seen that we can use the **inner product** between two signals $x(t)$ and $y(t)$ as a **measure of their similarity**:

$$\psi = \int_{-\infty}^{\infty} x(t)y^*(t)dt$$

Similarity measures are used to detect a signal buried in noise. F. ex. when a **pulse $x(t)$** is sent out into a medium, and a **reflected echo $y(t)$** is picked up. The reflected pulse will have a similar shape as the signal sent out but may be distorted due to noise.

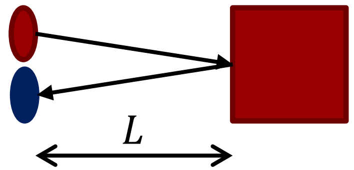
Another issue is that the reflected signal $y(t)$ will be delayed compared to the signal $x(t)$ sent out. This means that the two signals may not overlap in time, in which case their product will be zero in the expression showed at the top.

To circumvent this issue, a **correlation function** is defined where the similarity is determined for all possible delays. Here the independent variable t is the time shift of $x(t)$ compared to $y(t)$. At some range of t the shifted $x(t)$ will overlap $y(t)$ and produce a measure of similarity. The degree of similarity is expected to be largest when the two signals overlap completely.

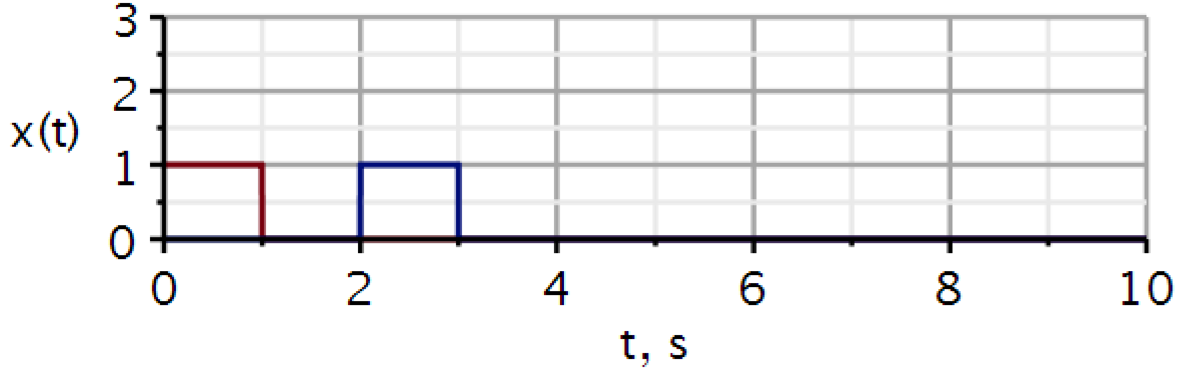
$$\psi_{xy}(t) = \int_{-\infty}^{\infty} x(\tau - t)y^*(\tau)d\tau$$

By looking for the similarity, we can find signals buried in noise.

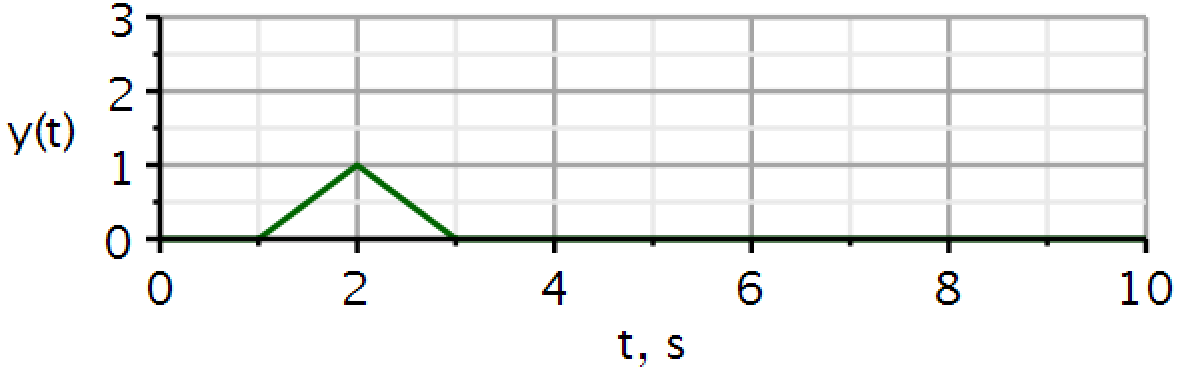
Correlation, an application of similarity measure



Here a **square pulse (red) is sent out** into a medium, and the **blue pulse is the reflected signal**. We see that the reflected signal takes 2 s to return to the wave detector/transmitter. If we know the wave velocity in the medium, we can calculate the distance between the transmitter and the reflecting object.



Calculating the correlation between the two pulses, we obtain the green curve. It starts at 1 s. This agrees with the observation that we have to shift the red curve 1 s before the red and blue curves start overlapping. At a time-shift of 2 s, the two curves are completely superimposed, giving the highest correlation. From thereon, the overlap will reduce, and the degree of correlation will decrease. Hence the green curve drops off for $t > 2s$.



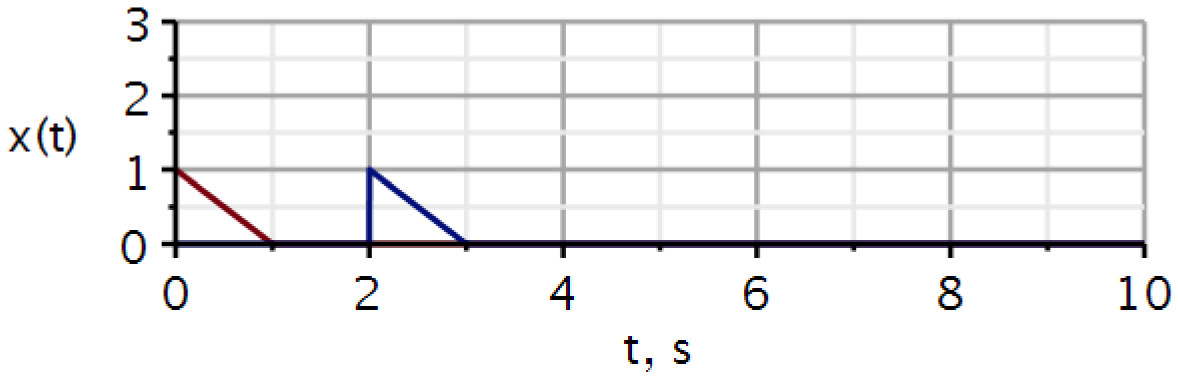
At a time-shift of 3 s, the red curve no longer overlaps the blue curve, and the correlation is zero.

$$L = v \cdot \frac{\Delta t}{2}$$

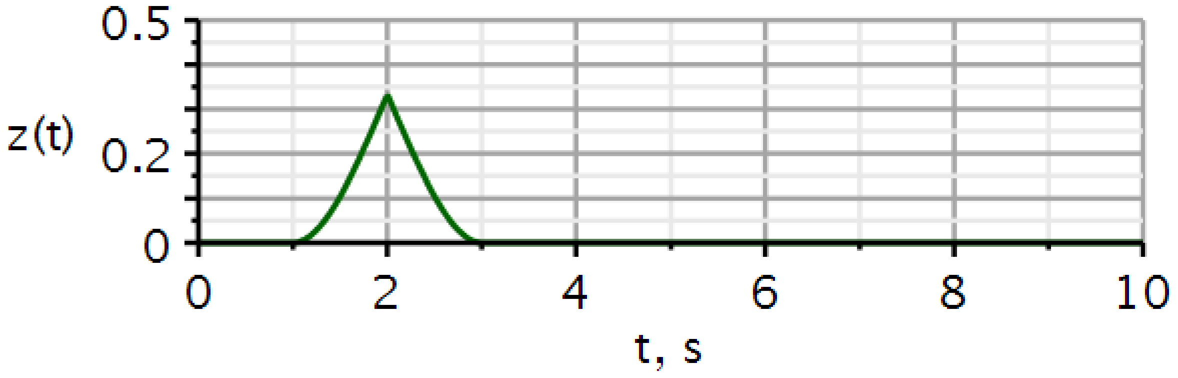
$$L = 1\,\frac{m}{s} \cdot \frac{2\,s}{2} = 1\,m$$

Correlation, an application of similarity measure

An example like the previous slide. Now the pulse sent out and received is a triangular waveform.



The correlation function indicates clearly that the reflected waveform is delayed 2 s.



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Lathi: Ch. 3.4 – 3.5 incl.

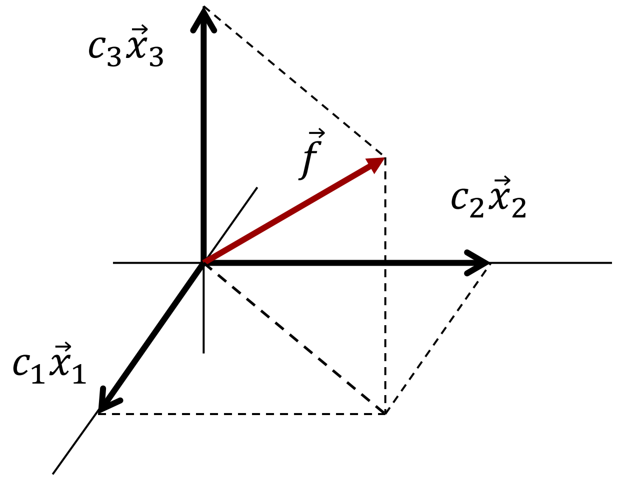
Fourier series

Video 1

Orthogonal expansion in vector space

Orthogonal expansion of vector $\vec{f}$ in terms of a complete orthogonal vector basis:

$$\vec{f} = c_1\vec{x}_1 + c_2\vec{x}_2 + c_3\vec{x}_3$$



Determination of component weights:

$$\vec{f}\cdot\vec{x}_1 = c_1(\vec{x}_1\cdot\vec{x}_1) + c_2(\vec{x}_2\cdot\vec{x}_1) + c_3(\vec{x}_3\cdot\vec{x}_1) \qquad c_1 = \frac{\vec{f}\cdot\vec{x}_1}{\vec{x}_1\cdot\vec{x}_1}$$

$$\vec{f}\cdot\vec{x}_2 = c_1(\vec{x}_1\cdot\vec{x}_2) + c_2(\vec{x}_2\cdot\vec{x}_2) + c_3(\vec{x}_3\cdot\vec{x}_2) \qquad c_2 = \frac{\vec{f}\cdot\vec{x}_2}{\vec{x}_2\cdot\vec{x}_2}$$

$$\vec{f}\cdot\vec{x}_3 = c_1(\vec{x}_1\cdot\vec{x}_3) + c_2(\vec{x}_2\cdot\vec{x}_3) + c_3(\vec{x}_3\cdot\vec{x}_3) \qquad c_3 = \frac{\vec{f}\cdot\vec{x}_3}{\vec{x}_3\cdot\vec{x}_3}$$

Orthogonal expansion of signals

We assume that we have a complete set of orthogonal signals (basis signals):

$$\{x_1(t), x_2(t), \dots\} \text{ where } \quad (x_m, x_n) \stackrel{\text{def}}{=} \int_{t_1}^{t_2} x_m(t)\, x_n^*(t) dt = \begin{cases} 0 & m \neq n \ (\perp) \\ E_n & m = n \ (\parallel) \end{cases}$$

By orthogonal expansion of a signal $f(t)$, we understand the representation of $f(t)$ as the weighted sum of orthogonal signals:

$$\begin{aligned} f(t) &= c_1 x_1(t) + c_2 x_2(t) + \cdots + c_n x_n(t) + \cdots \\ &= \sum_{n=1}^{\infty} c_n x_n(t), \qquad t_1 \leq t \leq t_2 \end{aligned}$$

Determination of component weights:

$$\begin{aligned} (f, x_m) = {} & c_1 (x_1, x_m) + c_2 (x_2, x_m) + \cdots \\ & + c_m (x_m, x_m) + \cdots \end{aligned}$$

$$\boxed{c_m = \frac{(f, x_m)}{(x_m, x_m)}}$$

Signal space

$$\boxed{c_m = \frac{\vec{f} \cdot \vec{x}_m}{\vec{x}_m \cdot \vec{x}_m}}$$

Vector space

Orthogonal expansion of signals

The equal sign here must be interpreted carefully.

$$f(t) = \sum_{n=1}^{\infty} c_n x_n(t)\,, \qquad t_1 \le t \le t_2$$

If we truncate the sum, the error function is the difference between the function and its expansion:

$$e(t) = f(t) - \sum_{n=1}^{N} c_n x_n(t)\,, \qquad t_1 \le t \le t_2$$

We recall that the coefficients of the expansion are such as to minimize the energy of the error signal:

$$E_e = \int_{t_1}^{t_2} |e(t)|^2 dt$$

As more terms are used in the expansion, the energy of the error will be reduced. However, it is the *integral* of a function squared that goes to zero.

$$E_e = \int_{t_1}^{t_2} |f(t)|^2 dt - \sum_{n=1}^{N} c_n^2 E_n$$

Appendix 3A

This means that the squared error function may be non-zero in a finite set of points, even if the integral is zero. Jump discontinuities are such points.

Parseval's theorem:

$$\int_{t_1}^{t_2} |f(t)|^2 dt = \sum_{n=1}^{\infty} c_n^2 E_n$$

For a complete basis set, energy is preserved.

Finality property

Because we are expanding the signal using an orthogonal basis set, we can add more terms to the series expansion without any effect on the previous terms already computed. So, if we want a more accurate approximation, we can just add more terms to the sum we have already computed. We do not have to recalculate all the coefficients.

We would need to do this if we changed an approximation from a first order polynomial fit to a second order polynomial fit.

$$f(t) \approx a_0 + a_1 t$$

$$f(t) \approx b_0 + b_1 t + b_2 t^2$$

$$a_0 \neq b_0, a_1 \neq b_1$$

$$f(t) = \sum_{n=1}^{\infty} c_n x_n(t), \qquad t_1 \leq t \leq t_2$$

$$f_{N_1}(t) = \sum_{n=1}^{N_1} c_n x_n(t), \qquad t_1 \leq t \leq t_2$$

$$f_{N_2}(t) = \sum_{n=1}^{N_2} c_n x_n(t), \qquad t_1 \leq t \leq t_2$$

$$f_{N_2}(t) = f_{N_1}(t) + \sum_{n=N_1+1}^{N_2} c_n x_n(t), t_1 \leq t \leq t_2$$

$$\boxed{c_n = \frac{1}{E_n} \int_{t_1}^{t_2} f(t)\, x_n^*(t) dt}$$

c_n depends only on x_n^*, not on other basis functions and not on other coefficients.

Trigonometric basis functions

It can be shown that:

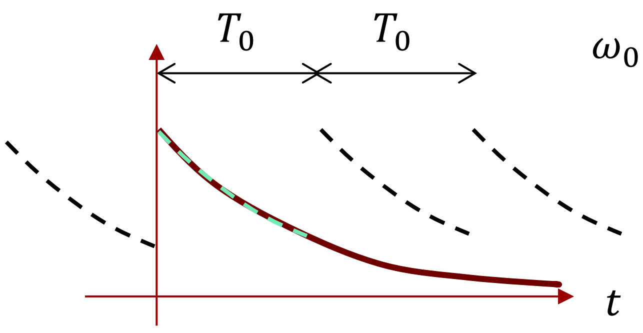
$$\int_{t_1}^{t_1+T_0} \cos(n\omega_0 t)\cos(m\omega_0 t)\,dt = \begin{cases} 0 & n \neq m\ (\perp) \\ \dfrac{T_0}{2} & n = m\ (\parallel) \end{cases}$$

$$\int_{t_1}^{t_1+T_0} \sin(n\omega_0 t)\sin(m\omega_0 t)\,dt = \begin{cases} 0 & n \neq m\ (\perp) \\ \dfrac{T_0}{2} & n = m\ (\parallel) \end{cases}$$

$$\int_{t_1}^{t_1+T_0} \sin(n\omega_0 t)\cos(m\omega_0 t)\,dt = 0\text{, for all n and m }(\perp)$$

The cosine and sine functions form **orthogonal sets**, and we can expand signals in such sets.

The angular frequency ω_0 is the fundamental frequency, having the time period T_0 corresponding to the time duration of the signal we want to expand.



$$\omega_0 = \frac{2\pi}{T_0}$$

$$\underbrace{f(t)}_{aperiodic} = \underbrace{a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)}_{periodic}, t_1 < t < t_1 + T_0$$

Here we expand $f(t)$ in the time interval from $t = 0$ to $t = T_0$.

Outside this interval, the two signals are not equal.

Trigonometric Fourier Series

By projecting the signal onto each basis function, we can calculate the coefficient for each basis function:

$$\boxed{c_m = \frac{(f, x_m)}{(x_m, x_m)}}$$

Fundamental frequency:

That frequency ω_0 which has the period T_0.

Harmonics:

All the other components. Their frequencies are integer multiples of the fundamental frequency. Hence, they all have an integer number of periods with the time interval T_0.

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t)\, dt : \text{average value within one period}$$

$$a_n = \frac{\int_{t_1}^{t_1+T_0} f(t) \cos(n\omega_0 t)\, dt}{\int_{t_1}^{t_1+T_0} \cos^2(n\omega_0 t)\, dt} = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos(n\omega_0 t)\, dt \,, n = 1,2,3, \ldots$$

$$b_n = \frac{\int_{t_1}^{t_1+T_0} f(t) \sin(n\omega_0 t)\, dt}{\int_{t_1}^{t_1+T_0} \sin^2(n\omega_0 t)\, dt} = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \sin(n\omega_0 t)\, dt \,, n = 1,2,3, \ldots$$

$$ratio\ of\ harmonic\ frequencies = \frac{n\ \omega_0}{m\ \omega_0} = \frac{n}{m} = a\ rational\ number$$

Compact trigonometric Fourier Series

The sum of a cosine and a sine with the same argument can be rewritten as a cosine with a phase angle θ:

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

Compact form

$$C_0 = a_0 \qquad C_n = \sqrt{a_n^2 + b_n^2} \qquad \theta_n = \tan^{-1}\left(-\frac{b_n}{a_n}\right)$$

We can verify that trigonometric Fourier Series are periodic:

$$f(t+T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0(t+T_0) + \theta_n)$$

$$f(t+T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + n\omega_0 T_0 + \theta_n)$$

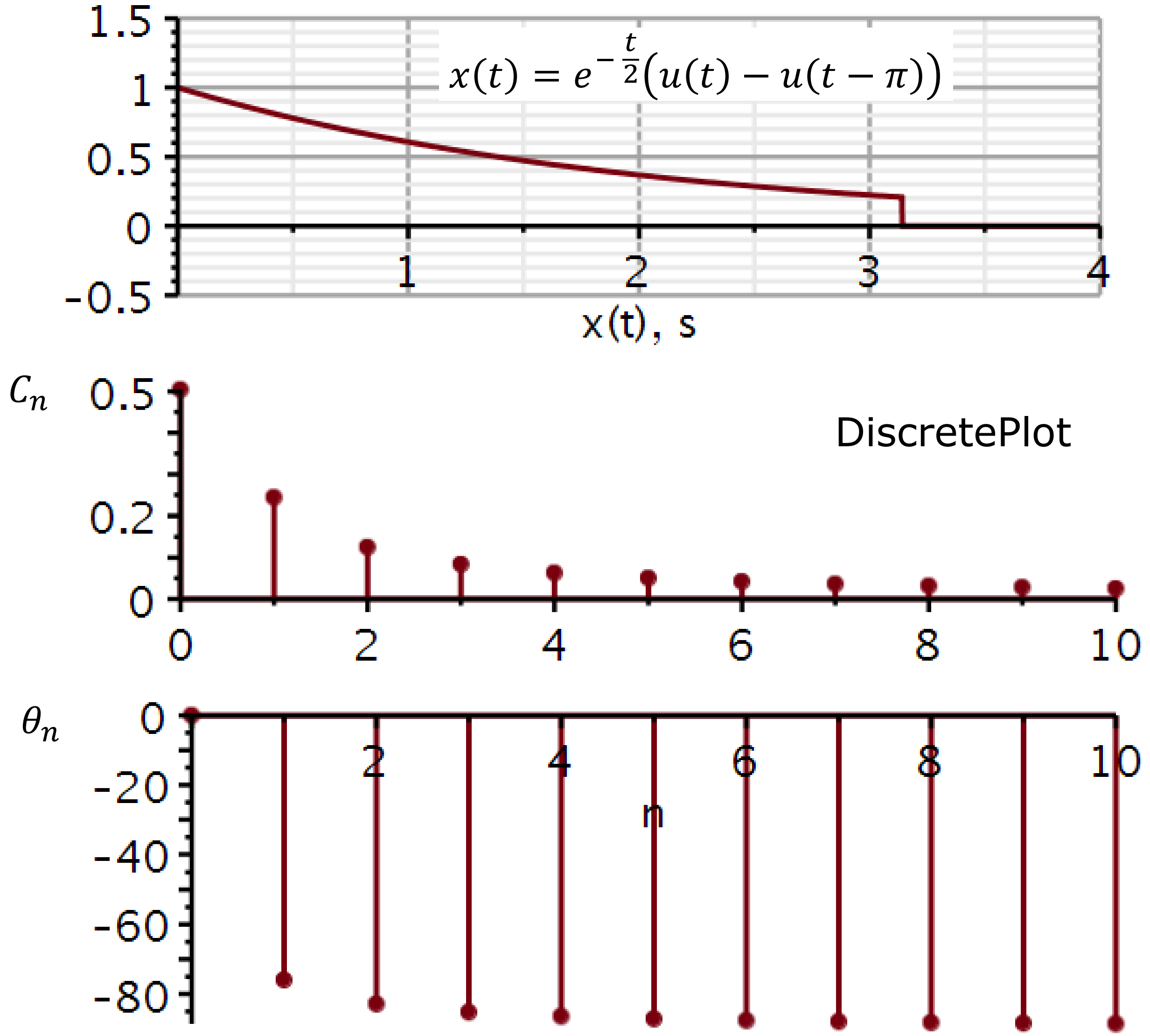
$$f(t+T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + n2\pi + \theta_n) = f(t)$$

Hence, even if $f(t)$ is aperiodic, the Fourier expansion over a time interval T_0 will be periodic with period T_0.

If a periodic signal $f(t)$ with period T_1 is expanded over the duration T_0 then the Fourier Series will have a period of T_0, while $f(t)$ has the period T_1. In this case, we would normally choose $T_0 = T_1$ such that the expansion has the same period as the periodic signal.

Trigonometric Fourier Series

Plotting the Fourier Series coefficients is an informative way of comparing signals. Plotting the a_n and b_n coefficients is not illuminating. However, plotting the coefficients of the *compact form* is very illuminating.



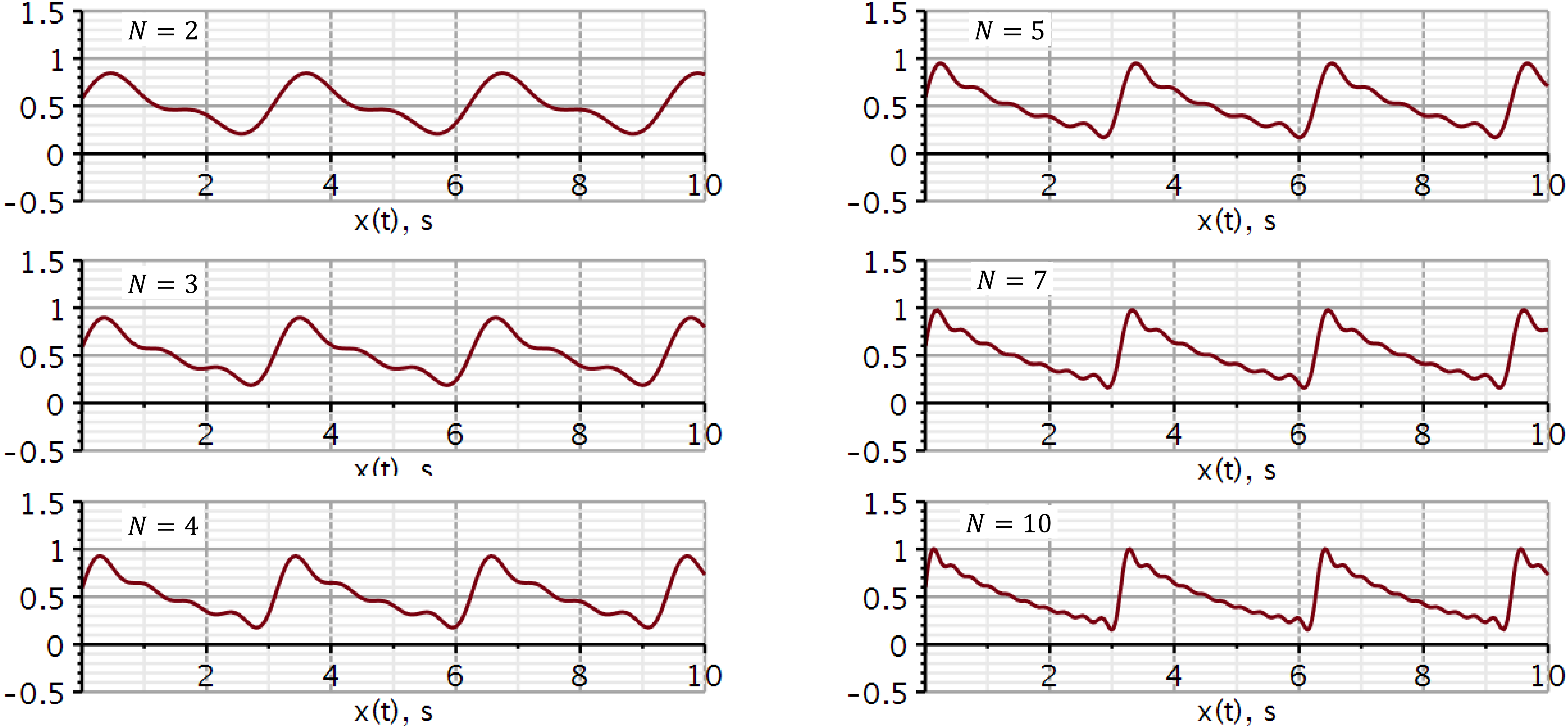
Example: Lathi example 3.5 (3.3 in old book)

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

The plot with C_n is the **magnitude plot** and the plot with θ_n is the **phase plot**.

The next slide shows the reconstruction of $x(t)$ using an increasing number of terms in the sum.

Reconstruction of exponential function using trigonometric Fourier Series expansion



Existence of Fourier series expansion

The existence of a Fourier Series expansion of a signal $f(t)$ requires that the integrals on the right converge.

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t)\, dt : \text{average value within one period}$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos n\omega_0 t \, dt \,, n = 1,2,3, \dots$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \sin n\omega_0 t \, dt \,, n = 1,2,3, \dots$$

A condition for convergence is that the signal has finite energy:

$$\int_{t_1}^{t_1+T_0} |f(t)|^2 dt < \infty$$

Weak Dirichlet condition

Must have a finite number of minima and maxima in one period and a finite number of discontinuities in one period.

$$\lim_{n\to\infty} a_n = 0$$

$$\lim_{n\to\infty} b_n = 0$$

Strong Dirichlet condition

Symmetry in even and odd signals

If $f(t)$ is an odd function in the period, then $a_0 = 0$.

$$a_0 = \frac{1}{T_0}\int_{t_1}^{t_1+T_0} f(t)\,dt = \begin{cases} 0 & f\ odd \\ \neq 0 & f\ even \end{cases}$$

A cosine is an even function. $odd \times even = odd$, hence $a_n = 0$ if $f(t)$ is an odd function.

$$a_n = \frac{2}{T_0}\int_{t_1}^{t_1+T_0} f(t)\cos n\omega_0 t\,dt = \begin{cases} 0 & f\ odd \\ \neq 0 & f\ even \end{cases}$$

A sine is an odd function. $even \times odd = odd$, hence $b_n = 0$ if $f(t)$ is an even function.

$$b_n = \frac{2}{T_0}\int_{t_1}^{t_1+T_0} f(t)\sin n\omega_0 t\,dt = \begin{cases} 0 & f\ even \\ \neq 0 & f\ odd \end{cases}$$

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- Trigonometric Fourier Series 3.4 – 3.5
  - Orthogonality of trigonometric basis functions
  - Trigonometric Fourier Series.
  - Compact trigonometric Fourier Series.
  - Conditions for existence.
  - Symmetry in even and odd signals
- Complex exponential Fourier Series 3.6
  - Orthogonality of complex exponential basis functions.
  - Determining coefficients
  - Symmetry properties
  - Negative frequencies
  - Power spectrum

Lathi: Ch. 3.6

Complex exponential Fourier series

Video 2

From trigonometric to complex exponential Fourier series

Using an Euler identity, we can rewrite the Fourier series expansion into one based on complex exponentials:

$$D_n = \frac{C_n}{2} e^{j\theta_n} = \frac{C_n}{2} (\cos\theta_n + j\sin\theta_n)$$

$$D_{-n} = \frac{C_n}{2} e^{-j\theta_n} = \frac{C_n}{2} (\cos\theta_n - j\sin\theta_n)$$

$$D_n = D_{-n}^*$$

We observe that the coefficients of the complex exponential Fourier series are complex valued.

We also observe that the magnitude of D_n is an even function of n and the phase an odd function of n.

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

$$\begin{aligned}
C_n \cos(n\omega_0 t + \theta_n) &= C_n \left(\frac{e^{j(n\omega_0 t + \theta_n)} + e^{-j(n\omega_0 t + \theta_n)}}{2} \right) \\
&= \frac{C_n}{2} \left(e^{j(n\omega_0 t + \theta_n)} + e^{-j(n\omega_0 t + \theta_n)} \right) \\
&= \frac{C_n}{2} e^{j\theta_n} e^{jn\omega_0 t} + \frac{C_n}{2} e^{-j\theta_n} e^{-jn\omega_0 t} \\
&= D_n e^{jn\omega_0 t} + D_{-n} e^{-jn\omega_0 t}
\end{aligned}$$

$$\begin{aligned}
x(t) &= C_0 + \sum_{n=1}^{\infty} \left(D_n e^{jn\omega_0 t} + D_{-n} e^{-jn\omega_0 t} \right) \\
&= \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}
\end{aligned}$$

Complex exponentials as basis signals

Using complex exponentials, the basis functions are:

$$\phi_n(t) = e^{jn\omega_0 t}, \qquad (n = 0, \pm 1, \pm 2, \dots)$$

We expand a periodic signal with period T_0:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, T_0 = 2\pi/\omega_0$$

The expansion is only useful if the basis signals are orthogonal:

$$I = \int_{t_1}^{t_1+T_0} e^{jm\omega_0 t} \left(e^{jn\omega_0 t}\right)^* dt = \int_{t_1}^{t_1+T_0} e^{j(m-n)\omega_0 t} dt$$

When $m = n$:

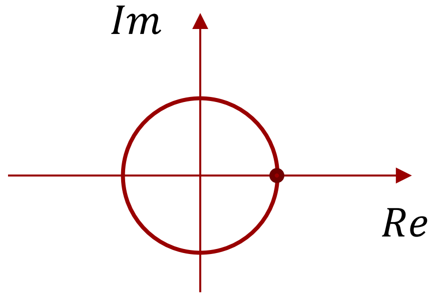
$$I = \int_{t_1}^{t_1+T_0} e^0 dt = [t]_{t_1}^{t_1+T_0} = T_0$$

When $m \neq n$:

$$I = \frac{1}{j(m-n)\omega_0} e^{j(m-n)\omega_0 t}\Big|_{t_1}^{t_1+T_0}$$

$$= \frac{1}{j(m-n)\omega_0} e^{j(m-n)\omega_0 t_1}\left[e^{j(m-n)\omega_0 T_0} - 1\right]$$

$$= \frac{1}{j(m-n)\omega_0} e^{j(m-n)\omega_0 t_1}\left[e^{j(m-n)\omega_0 \frac{2\pi}{\omega_0}} - 1\right] = 0$$



Why?

Determining expansion coefficients

The complex exponentials form a set of **orthogonal basis signals** over the time-period T_0.

$$\int_{t_1}^{t_1+T_0} e^{jm\omega_0 t}\left(e^{jn\omega_0 t}\right)^* dt = \begin{cases} 0 & m \neq n\ (\perp) \\ T_0 & m = n\ (\parallel) \end{cases}$$

Fourier series expansion in complex exponentials:

$$x(t) = \sum_{m=-\infty}^{\infty} D_m e^{jm\omega_0 t}, t_1 \leq t \leq t_1 + T_0, T_0 = \frac{2\pi}{\omega_0}$$

To determine the weights D_n, we form the inner product with the n'th basis signal:

$$\left(x, e^{-jn\omega_0 t}\right) = \int_{t_1}^{t_1+T_0} x(t) e^{-jn\omega_0 t} dt$$

$$\left(e^{jn\omega_0 t}\right)^* = e^{-jn\omega_0 t}$$

$$= \int_{t_1}^{t_1+T_0} \sum_{m=-\infty}^{\infty} D_m e^{jm\omega_0 t}\, e^{-jn\omega_0 t} dt$$

$$= \sum_{m=-\infty}^{\infty} D_m \int_{t_1}^{t_1+T_0} e^{jm\omega_0 t} e^{-jn\omega_0 t} dt = D_n T_0$$

We have seen that the coefficients D_n can be obtained from C_n from the trigonometric Fourier series. But we can also compute it using the method of projection as shown here.

Coefficients of complex exponential Fourier series:

$$\boxed{D_n = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} x(t) e^{-jn\omega_0 t} dt}$$

Summary of Fourier Series expansion

- Fourier synthesis
  - We construct a signal from orthogonal basis signals

$$\boxed{x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, t_1 \leq t \leq t_1 + T_0, T_0 = \frac{2\pi}{\omega_0}}$$

- Fourier analysis
  - We break the signal up into its components and determine the complex-valued coefficient of each.

$$\boxed{D_n = \frac{1}{T_0}\int_{t_1}^{t_1+T_0} x(t)e^{-jn\omega_0 t}dt}$$

Symmetries for real-valued signals

To elucidate the symmetry of the coefficients D_n we expand the **REAL-valued** signal $x(t)$ into its **even** and **odd** components.

We observe, that

- The real part of D_n originates from the even part of the signal.
- The imaginary part of D_n originates from the odd component.

Hence

- If the signal is even, D_n has zero imaginary part.
- If the signal is odd, D_n has zero real part.

$$D_n = \frac{1}{T_0}\int_{t_1}^{t_1+T_0} x(t)e^{-jn\omega_0 t}dt$$

$$D_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t) + x_o(t)\big)\big(\cos(n\omega_0 t) - j\sin(n\omega_0 t)\big)dt$$

$$= \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t)\big)\big(\cos(n\omega_0 t)\big)dt \qquad \text{real}$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t)\big)\big(-j\sin(n\omega_0 t)\big)dt \qquad = 0$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_o(t)\big)\big(\cos(n\omega_0 t)\big)dt \qquad = 0$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_o(t)\big)\big(-j\sin(n\omega_0 t)\big)dt \qquad \text{imaginary}$$

Symmetries for real-valued signals

We furthermore observe, that

- The real part of D_n is an even function of n.
- The imaginary part of D_n is an odd function of n.

This means that:

- The magnitude of D_n is an even function of n.
- The phase of D_n is an odd function of n.

For complex-valued signals twice as many combinations apply.

Always have these symmetries in mind when forecasting what to expect when you calculate Fourier series coefficients.

$$D_n = \frac{1}{T_0}\int_{t_1}^{t_1+T_0} x(t)e^{-jn\omega_0 t}dt$$

$$D_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t) + x_o(t)\big)\big(\cos(n\omega_0 t) - j\sin(n\omega_0 t)\big)dt$$

$$= \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t)\big)\big(\cos(n\omega_0 t)\big)dt \qquad \text{even in } n$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_e(t)\big)\big(-j\sin(n\omega_0 t)\big)dt \qquad = 0$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_o(t)\big)\big(\cos(n\omega_0 t)\big)dt \qquad = 0$$

$$+ \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \big(x_o(t)\big)\big(-j\sin(n\omega_0 t)\big)dt \qquad \text{odd in } n$$

Examples

We will use Maple to carry out a Fourier Series expansion and reconstruction of a periodic square pulse signal.

Including the **DynamicSystems** package allows us to use certain predefined signals, here the Square() signal.

$restart$

$with(DynamicSystems):$

$with(plots):$

Time domain signal

$$x := t \rightarrow Square\left(1, 2\pi \cdot 1, \frac{1}{2}, -\frac{1}{4}\right):$$

Height

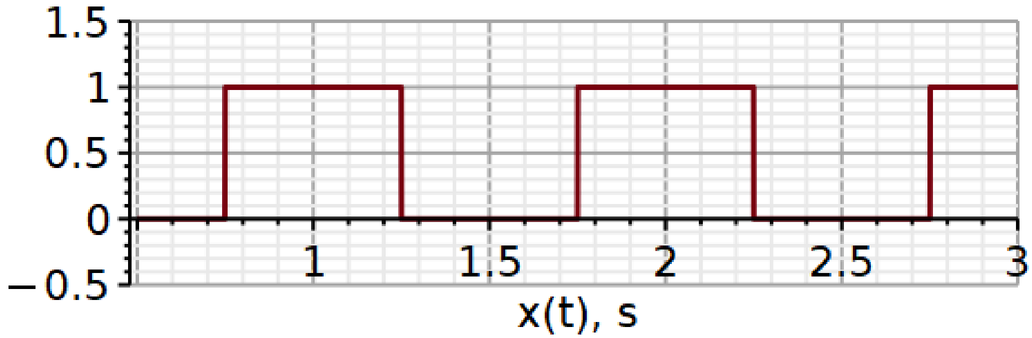
Frequency

Duty cycle

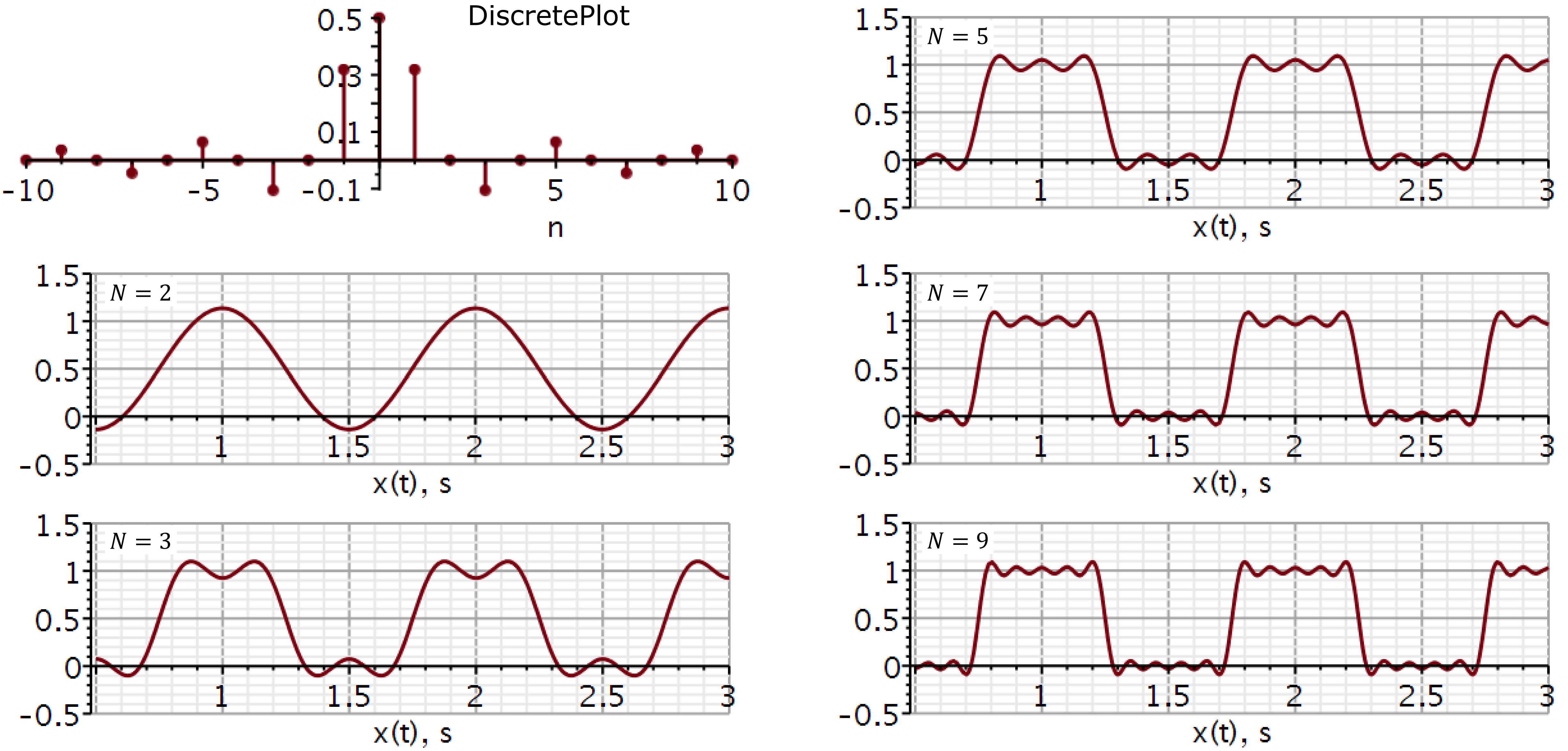
Delay

Plotting signal

$plot(x(t), t = 0.5..3.0, -0.5..1.5, thickness = 3, axesfont = [Helvetica, "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = ["x(t), s", " "], labelfont = [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200])$



Examples



Power spectrum

A cosine with amplitude A has the power $A^2/2$.

We can extend this to the trigonometric Fourier series:

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

$$P_f = C_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} C_n^2$$

We can extend this to the complex Fourier series:

$$f(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \qquad D_n = \frac{C_n}{2}\, e^{j\theta_n}$$

$$P_f = D_0^2 + 2\sum_{n=1}^{\infty} |D_n|^2 \qquad |D_n|^2 = \frac{C_n^2}{4}$$

In a **power spectrum plot**, usually only positive frequency stems are plotted. However, the magnitude of each stem is doubled (except DC) to include the power of the negative frequencies.

Learning priorities from Problem Solving:

1. Why this method?
2. When is this method valid?
3. How is this method used?
4. What is the result?
5. How do I validate the result?

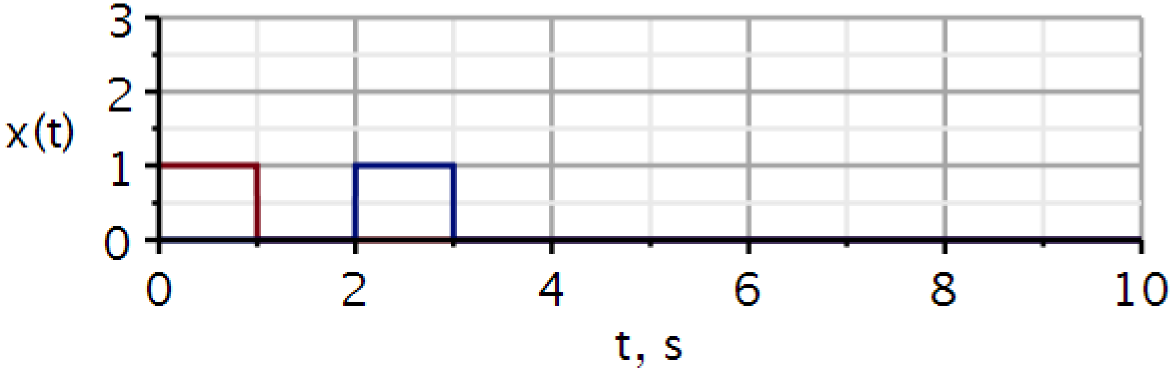
Problems

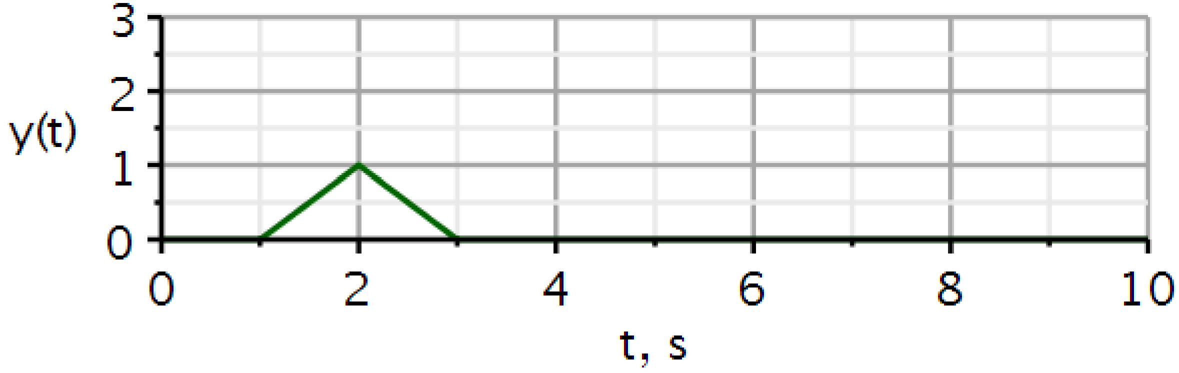
Problem 1

Here a square pulse (red) is sent out into a medium, and the blue pulse is the reflected signal. We see that the reflected signal takes 2 s to return to the wave detector/transmitter. If we know the wave velocity in the medium, we can calculate the distance between the transmitter and the reflecting object.

Use Maple and:

1. Define the two signals. Hint: use combinations of step functions.
2. Plot them in one graph.
3. Define a correlation integral.
4. Use it to plot the correlation $y(t)$.





If needed, see hints on next slide.

Problem 1 Hint

$v := t \rightarrow \text{Heaviside}(t)$:

$corr := (x, y) \rightarrow t \rightarrow int(x(\tau - t) * y(\tau), \tau = -10..10);$

$$corr := (x, y) \mapsto t \mapsto \int_{-10}^{10} x(\tau - t) \cdot y(\tau) \, d\tau$$

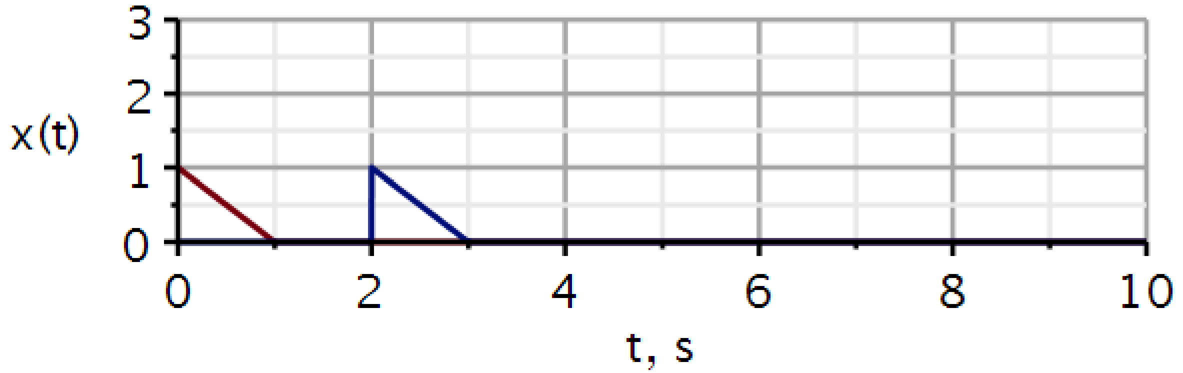
x: one signal
y: another signal
t: the time delay of x to make x overlap y
Fixed integration interval. Must be increased if signals extend beyond that.

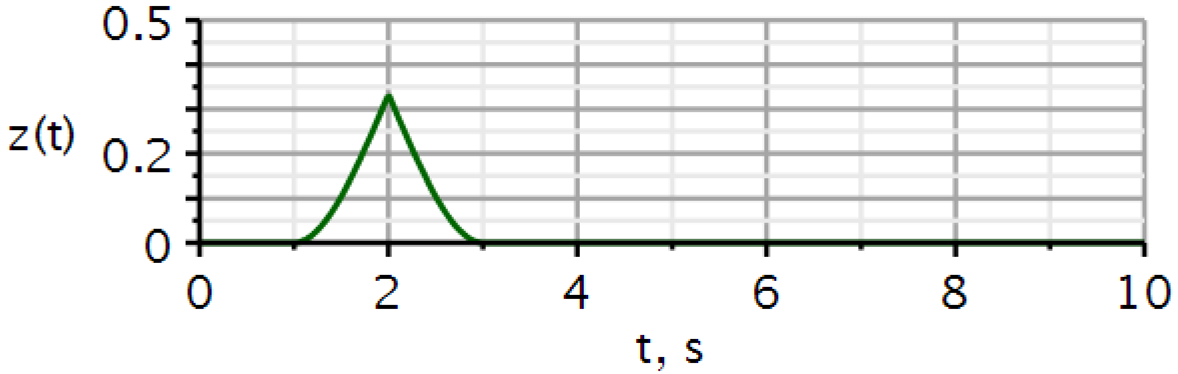
x1: signal sent out
y1: signal reflected
z1: correlation signal indicating similarity and time delay between the two signals

$x1 := t \rightarrow (v(t) - v(t-1))$: *# signal sent out*
$y1 := t \rightarrow (v(t-2) - v(t-3))$: *# signal returned*
$z1 := t \rightarrow corr(t \rightarrow x1(t), t \rightarrow y1(t))(t)$: *# correlation signal*

Problem 2

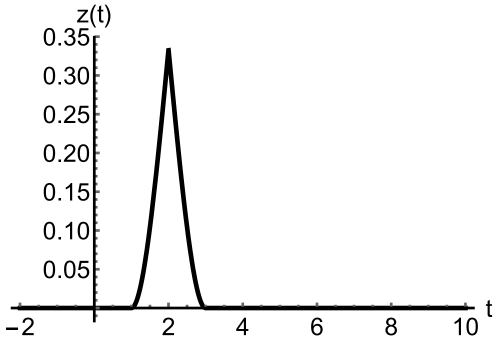
An example like the previous slide. Now the pulse sent out and received is a triangular waveform.





Use Maple and:

1. Define the two signals.
2. Plot them in one graph.
3. Define a correlation integral.
4. Use it to plot the correlation $z(t)$.



In Mathematica

Problem 3

Example: Lathi example 3.5 (3.3 in old book)

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

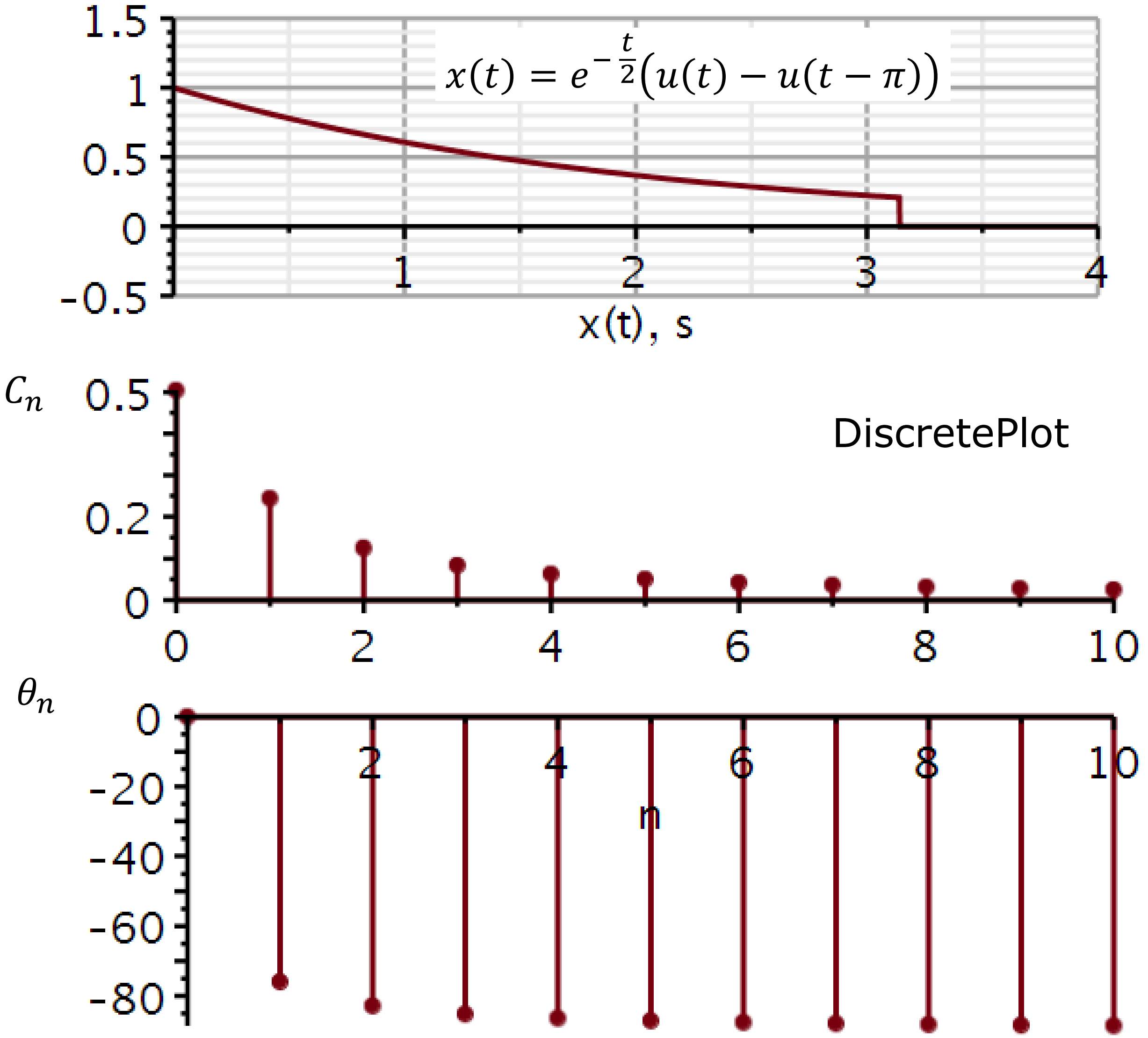
Use Maple to reproduce the plots shown here using the trigonometric Fourier series. Validate that your calculated coefficients agree with those given in example 3.5 (3.3).

Figure out how to use DiscretePlot. You need the DynamicSystems package.

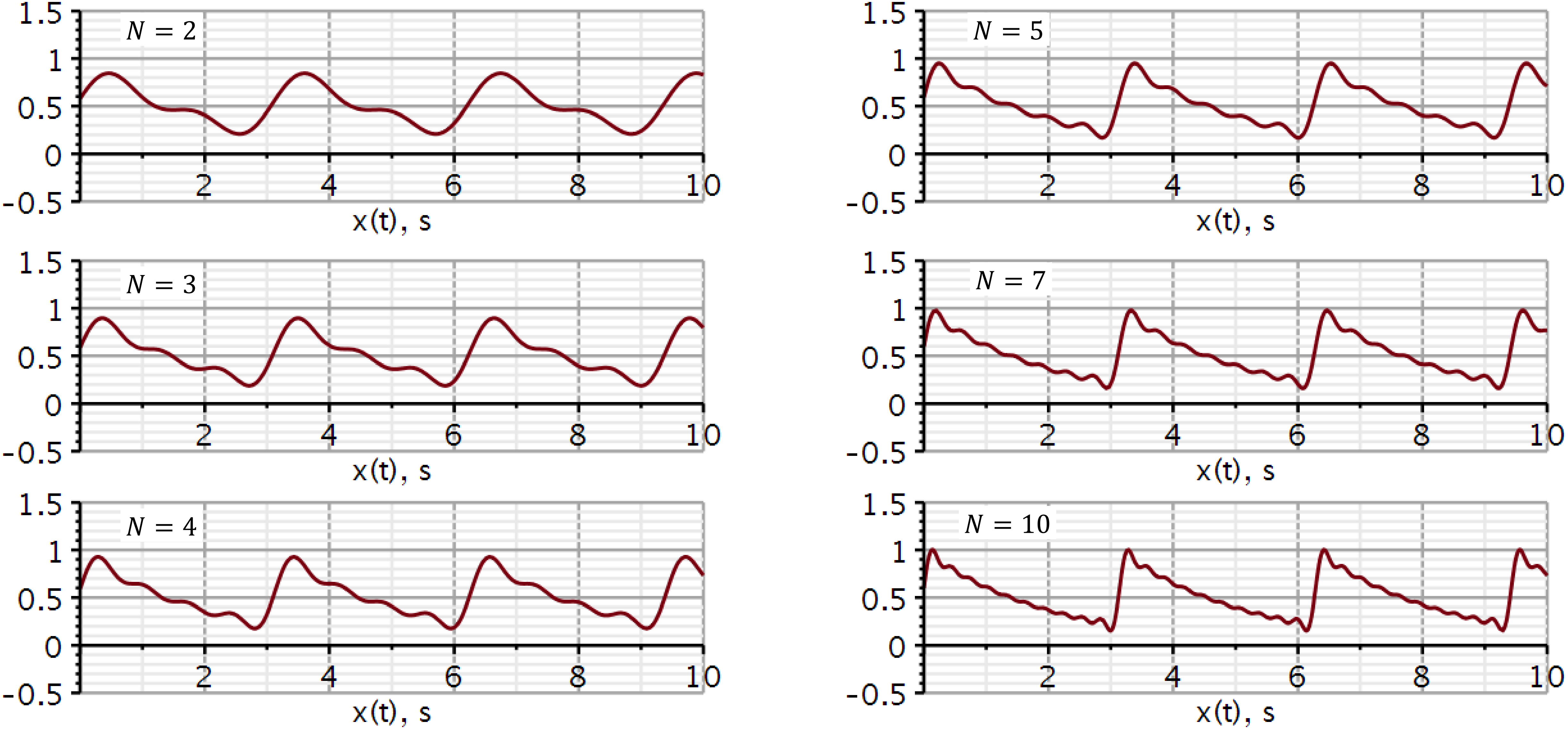
I organized my data using *Vector.*

Also do the reconstructions shown on the next slide.

See hints below.



Problem 3



Problem 3: Solution

We need to calculate the Fourier series coefficients in Maple. Here is defined three functions that do just that:

Calculation of Fourier Series coefficients:

$$a0 := (n, t1, T0) \rightarrow \frac{1}{T0} \cdot \int_{t1}^{t1+T0} x(t)\,\mathrm{d}t :$$

$$a := (n, t1, T0) \rightarrow \frac{2}{T0} \cdot \int_{t1}^{t1+T0} x(t) \cdot \cos\left(n \cdot \frac{2 \cdot \pi}{T0} \cdot t\right) \mathrm{d}t :$$

$$b := (n, t1, T0) \rightarrow \frac{2}{T0} \cdot \int_{t1}^{t1+T0} x(t) \cdot \sin\left(n \cdot \frac{2 \cdot \pi}{T0} \cdot t\right) \mathrm{d}t :$$

n: index of Fourier Series coefficient
t1: start time a signal period
T0: Period duration
x(t): The periodic time-signal to be decomposed in a Fourier series. Must be defined before calling this function.

Problem 3: Solution

We need to define the signal:

We cannot call the functions on the previous slide until after the signal is defined.

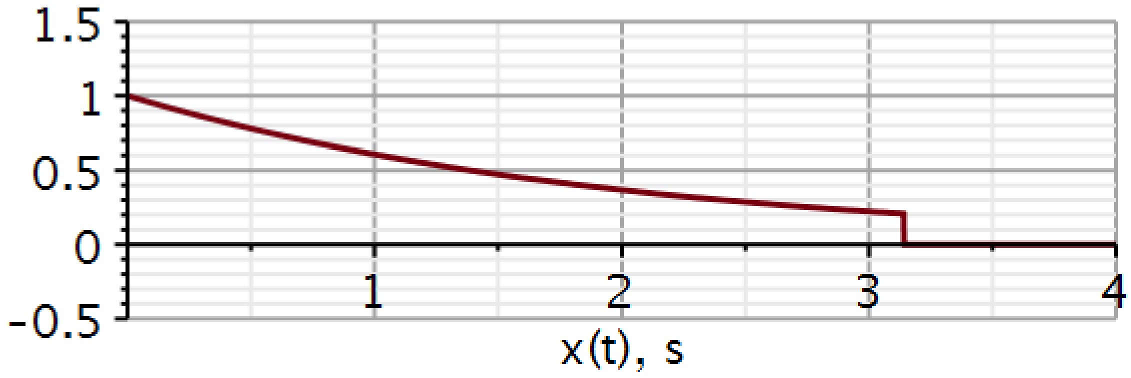
Time domain signal

$$x := t \rightarrow \mathrm{e}^{-\frac{t}{2}} \cdot \left(v(t) - v(t-\pi) \right) :$$

$$T0 := \pi :$$

Plotting signal

$plot(x(t), t = 0..4, -0.5..1.5, thickness = 3, axesfont = [Helvetica, \text{"roman"}, 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = [\text{"x(t), s"}, \text{" "}], labelfont = [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200])$



Problem 3: Solution

When you have defined the integrals, test the code by calculating and printing out a_0, a_1, b_1 for example 3.5 (3.3) in Lathi. This is my validation.

Validating function

$$a0 := evalf(a0(0, 0, T0))$$

$$a0 := 0.5042795237$$

$$a1 := evalf(a(1, 0, T0))$$

$$a1 := 0.05932700278$$

$$b1 := evalf(b(1, 0, T0))$$

$$b1 := 0.2373080111$$

$$C1 := \sqrt{a1^2 + b1^2}$$

$$C1 := 0.2446114989$$

$$\theta 1 := \frac{180}{\pi} \arctan\left(-\frac{b1}{a1}\right)$$

$$\theta 1 := -75.96375653$$

We need many coefficients, so we store them in Maple vectors (see next slide).

Problem 3: Solution

Index vector $\quad n \quad a_n \quad b_n$

$T := Vector(11, n \mapsto n - 1)$ $\quad A := Vector(10, n \rightarrow evalf(a(n, 0, T0), 5))$ $\quad B := Vector(10, n \rightarrow evalf(b(n, 0, T0), 5))$

$$T := \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ \vdots \end{bmatrix} \qquad A := \begin{bmatrix} 0.059329 \\ 0.015516 \\ 0.0069555 \\ 0.0039244 \\ 0.0025151 \\ 0.0017479 \\ 0.0012848 \\ 0.00098395 \\ 0.00077760 \\ 0.00062995 \end{bmatrix} \qquad B := \begin{bmatrix} 0.23731 \\ 0.12413 \\ 0.083464 \\ 0.062793 \\ 0.050304 \\ 0.041951 \\ 0.035975 \\ 0.031487 \\ 0.027995 \\ 0.025198 \end{bmatrix}$$

11 element Vector[column]

Be aware that the first element in an array in Maple starts at index 1.

Problem 3: Solution

When I have my a_n and b_n coefficients saved in vectors $A[n]$ and $B[n]$, I need to make the $C[n]$ and $\theta[n]$ vectors.

I start by making a C Vector with 11 elements, all with the value $a0$.

Then I replace elements 2 to 11 with their proper values:

$$C := Vector(11, init, a0)$$

Overwriting all values except a1 with C[n] values, n>1.

$$C[2..11] := Vector\left(10, n \to evalf\left(\sqrt{(A[n])^2 + (B[n])^2}, 3\right)\right)$$

Validating the content of C vector

C

$$\begin{bmatrix} 0.5042795237 \\ 0.244 \\ 0.125 \\ 0.0838 \\ 0.0629 \\ 0.0504 \\ 0.0420 \\ 0.0361 \\ 0.0315 \\ 0.0280 \\ \vdots \end{bmatrix}$$

11 element Vector[column]

Problem 3: Solution

A similar technique is used to build the $\theta[n]$ vector.

Initializing vector for phase angles.

$$\theta := Vector(11, init, 0)$$

$$\theta := \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{bmatrix}$$

11 element Vector[column]

Filling in all values for n>1

$$\theta[2..11] := Vector\left(10, n \rightarrow evalf\left(\frac{180}{\pi} \cdot \arctan\left(-\frac{B[n]}{A[n]}\right)\right)\right)$$

$$\theta_{2..11} := \begin{bmatrix} -75.96341568 \\ -82.87509722 \\ -85.23624467 \\ -86.42381332 \\ -87.13770836 \\ -87.61413550 \\ -87.95462557 \\ -88.21012343 \\ -88.40893921 \\ -88.56790378 \end{bmatrix}$$

Problem 3: Solution

Here are the two plot statements I use to plot the amplitudes and phase angles as stem-plots.

I used *Vectors* to hold x-axis and y-axis data pairs. Hence T, C, θ are all Maple Vectors.

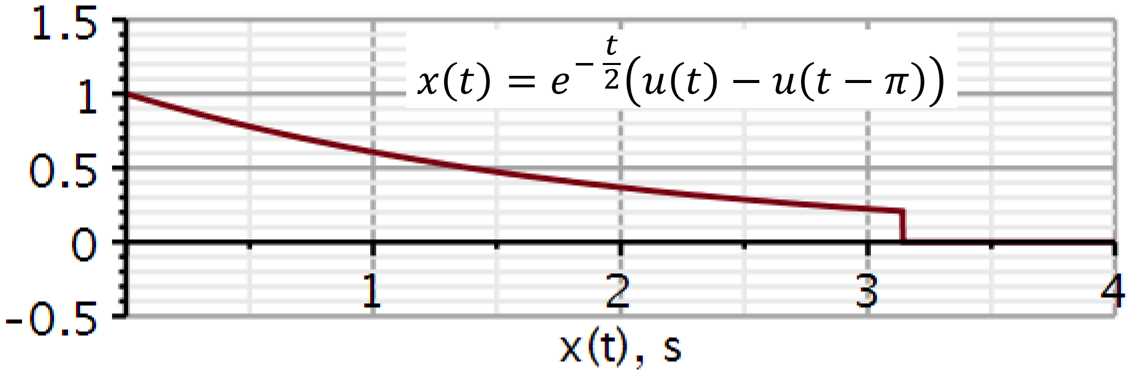
I don't know if that is the only way, but it worked for me.

$$DiscretePlot(T, C, style = stem, \; thickness = 3, symbol = solidcircle, symbolsize = 18, axesfont = [Helvetica, roman, 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = [\text{"n"}, \text{" "}], labelfont = [Helvetica, roman, 18], size = [600, 200])$$

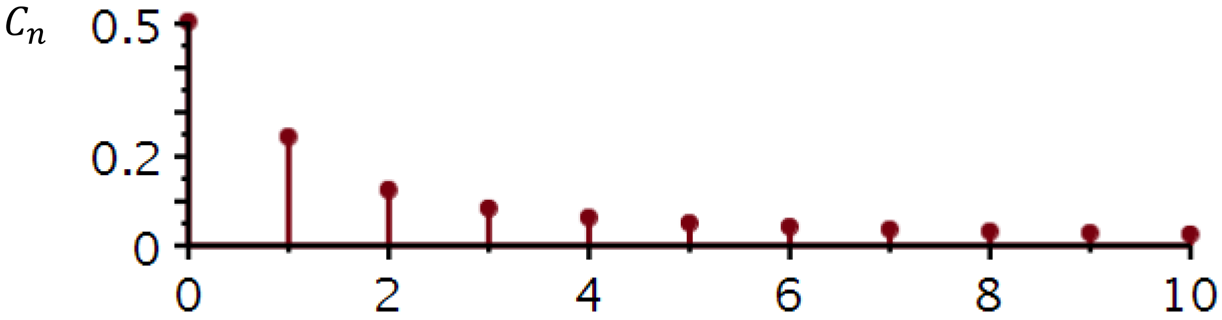
$$DiscretePlot(T, \theta, style = stem, \; thickness = 3, symbol = solidcircle, symbolsize = 18, axesfont = [Helvetica, roman, 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = [\text{"n"}, \text{" "}], labelfont = [Helvetica, roman, 18], size = [600, 200])$$

Problem 3: Solution

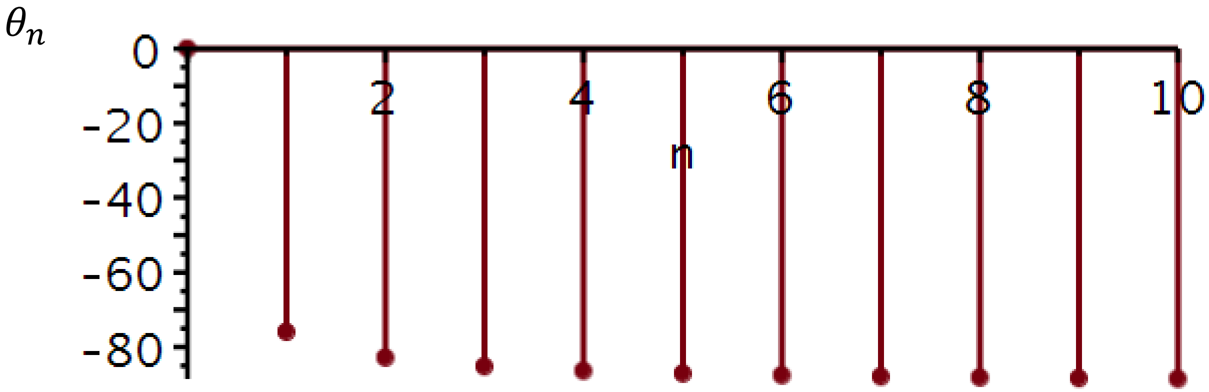
DiscretePlot



Amplitude spectrum:



Phase spectrum:



Problem 3: Solution

Here is the Maple code to reconstruct the time signal from its Fourier coefficients:

Synthesis of continuous-time signal from Fourier series coefficients

The coeffcients are stored in array A and B starting with index = 1.

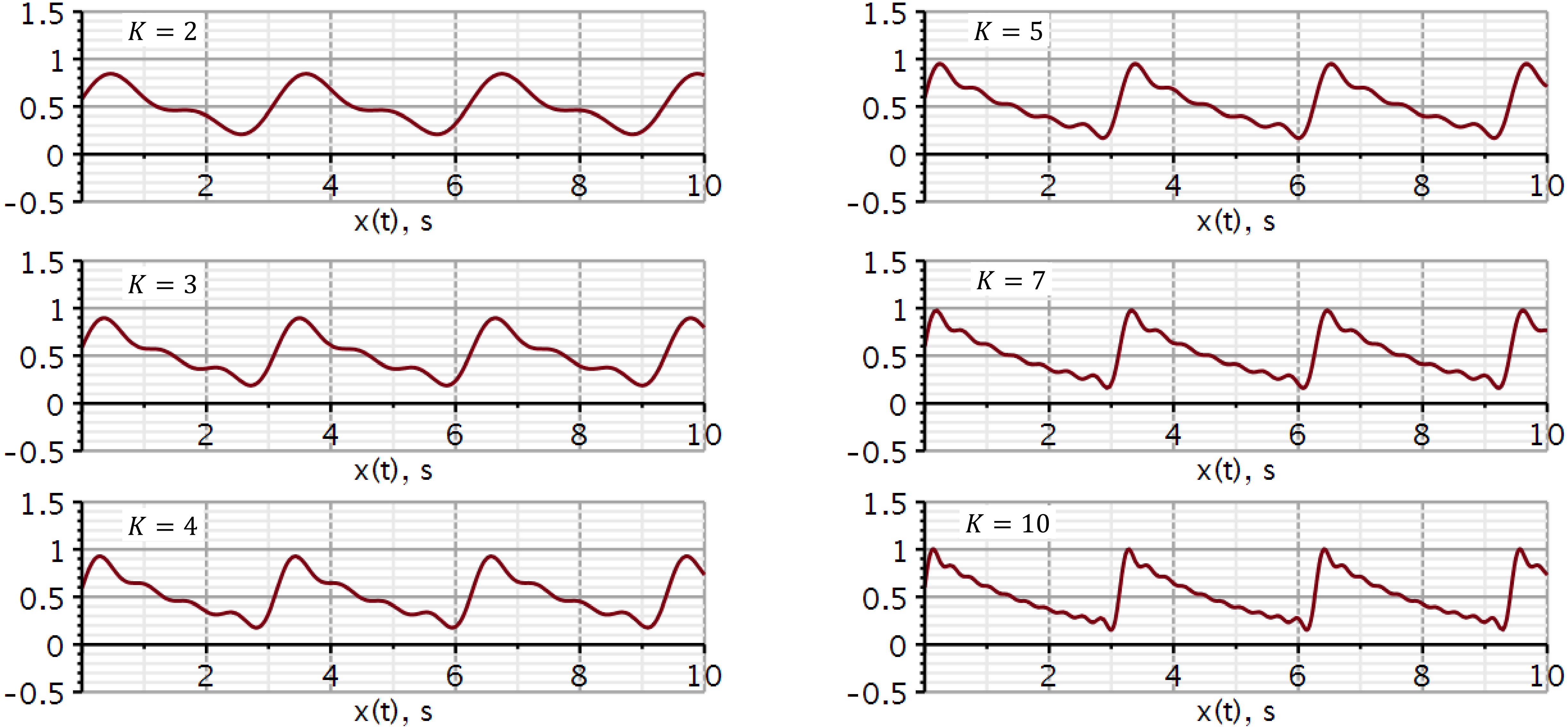
$$y := (a0, A, B, T0, K, t) \rightarrow a0 + \sum_{n=1}^{K} \left('A[n]' \cdot \cos\left(n \cdot \frac{2 \cdot \pi}{T0} \cdot t\right)\right) + \sum_{n=1}^{K} \left('B[n]' \cdot \sin\left(n \cdot \frac{2 \cdot \pi}{T0} \cdot t\right)\right) :$$

A: array of Fourier coefficients. Must be with the appostrophs ('). Some kind of delay.
B: array of Fourier coefficients. Must be with the appostrophs ('). Some kind of delay.
T0: Time period of periodic signal.
K: The number of coefficients we want to include in the reconstruction (synthesis) of the time-signal.

$K = 1$

$plot(y(a0, A, B, T0, 1, t), t = 0..10.0, -0.5..1.5, thickness = 3, font = [\text{"Helvetica"}, \text{"roman"}, 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = [\text{"x(t), s"}, \text{" "}], labelfont = [\text{"HELVETICA"}, 18], numpoints = 100, gridlines, size = [600, 200])$

Problem 3: Solution



Problem 4

Use Maple to carry out a complex exponential Fourier Series expansion and reconstruction of a periodic square pulse signal.

Reconstruct the plots on this and the next slide. Note that the plots on the next slide have been copied and pasted into this layout. Don't bother to organize plots in a grid.

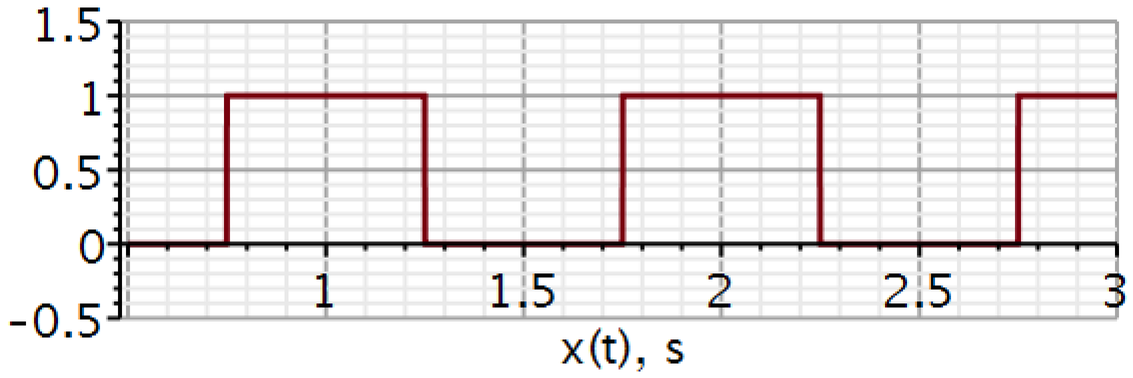
restart
with(*DynamicSystems*) :
with(*plots*) :

Time domain signal

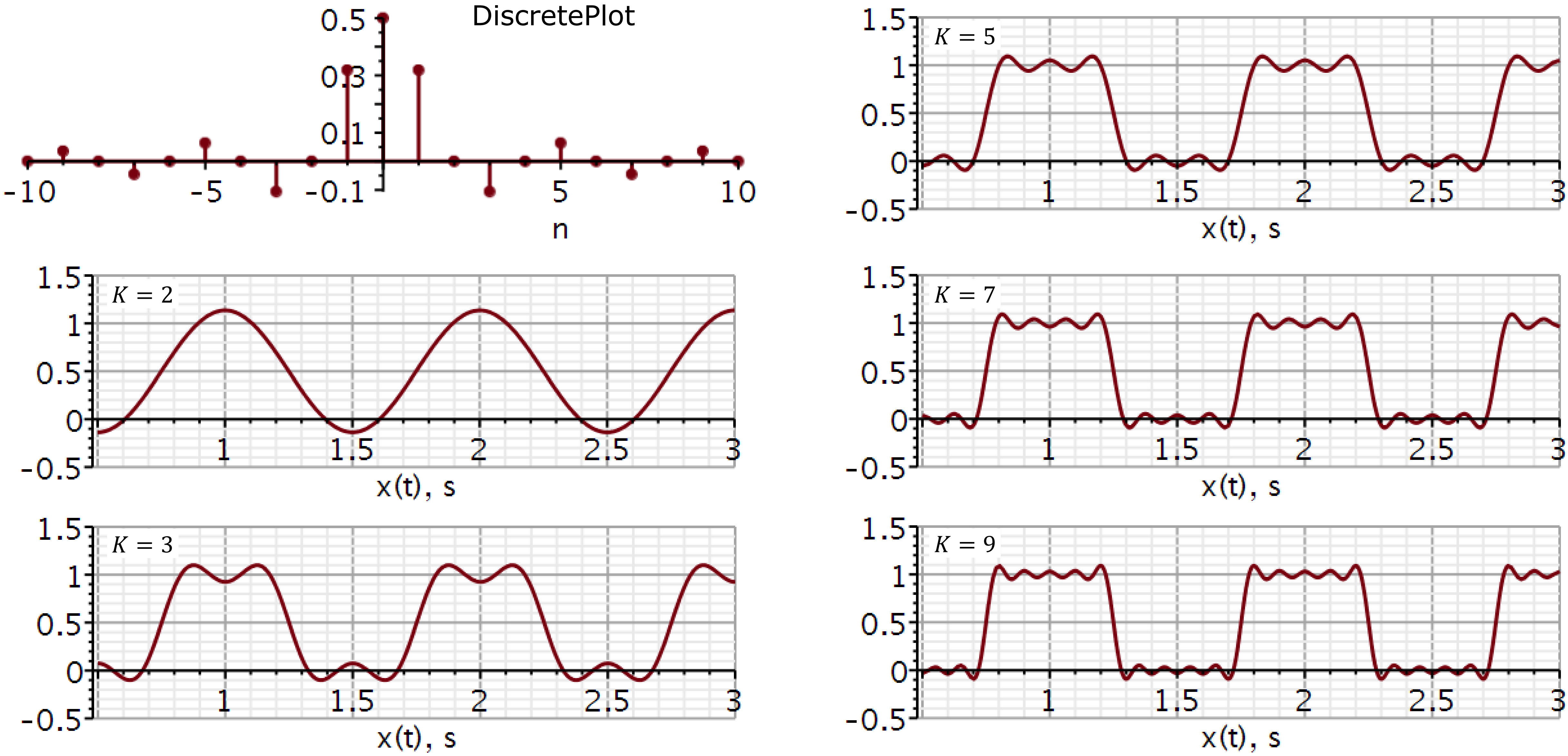
$$x := t \rightarrow Square\left(1, 2\pi \cdot 1, \frac{1}{2}, -\frac{1}{4}\right) :$$

Plotting signal

plot(*x*(*t*), *t* = 0.5 ..3.0, −0.5 ..1.5, *thickness* = 3, *axesfont* = [*Helvetica*, "roman", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*thickness* = 2.5], *labels* = ["x(t), s" , " "], *labelfont* = [*Helvetica*, 18], *numpoints* = 100, *gridlines*, *size* = [600, 200])



Problem 4


DiscretePlot
0.5
0.3
0.1
-0.1
-10
-5
5
10
n
K = 2
K = 3
K = 5
K = 7
K = 9
1.5
1
0.5
0
-0.5
1
1.5
2
2.5
3
x(t), s

Problem 4: Solution

Synthesis of continuous-time signal from Fourier series coefficients

The summation is symmetric about zero, however, the coeffcients are stored in array A starting witn index = 1.

$$y := (A, T0, M, K, t) \rightarrow \sum_{n=-K}^{K} \left('A[n+M+1]' \cdot e^{\frac{j \cdot 2 \cdot \pi \cdot n \cdot t}{T0}} \right) :$$

A: array of Fourier coefficients. Must be with the appostrophs ('). Some kind of delay.
T0: Time period of periodic signal.
M: Total number of (positive) coefficients calculated. If A holds 21 values (10 negative, 1 zero and 10 positive) then M=10.
K: The number of coefficients we want to include in the reconstruction (synthesis) of the time-signal. K can be less than M to see a worse reconstruction, but cannot be larger than M.

While n runs from $-K$ to K, the coefficients are stored in A from index 1 to index $K + M + 1$. Workout the position of coefficients in the vector by doing example numbers on a piece of paper.

Problem 4: Solution

You will have to figure out yourselves how to define the integrals in Maple to compute the coefficients C_n.

When you have defined the integrals test the code by calculating and printing out $C[0 \dots 3]$. This is my validation.

Validating function

$$evalf\left(C\left(0, \frac{3}{4}, 1\right)\right)$$

$$0.5000000000$$

$$evalf\left(C\left(1, \frac{3}{4}, 1\right)\right)$$

$$0.3183098862 + 1.309745927\ 10^{-16}\ \mathrm{I}$$

$$evalf\left(C\left(2, \frac{3}{4}, 1\right)\right)$$

$$2.185976251\ 10^{-17} - 4.882812500\ 10^{-17}\ \mathrm{I}$$

$$evalf\left(C\left(3, \frac{3}{4}, 1\right)\right)$$

$$-0.1061032954 - 6.938893904\ 10^{-17}\ \mathrm{I}$$

Problem 4:

Why should we not bother to plot the phase angles in this case?